



Immediate Effects of Dry Needling and Myofascial Release on Local and Widespread Pressure Pain Threshold in Individuals With Active Upper Trapezius Trigger Points: A Randomized Clinical Trial

Fábio F. Stieven, PT, DC, PhD,^{a, b} Giovanni E. Ferreira, PT,^{c, d} Francisco X. de Araújo, PT, PhD,^e Rodrigo F. Angellos, MsC,^f Marcelo F. Silva, PhD,^a and Luis Henrique T. da Rosa, PhD^a

ABSTRACT

Objective: This study sought to compare the immediate effects of a single session of dry needling (DN), myofascial release (MR), and sham DN on pressure pain threshold (PPT) and neck pain intensity in individuals with chronic neck pain.

Methods: This was a randomized trial with a blinded outcome assessor. Forty-four individuals with chronic neck pain and unilateral myofascial trigger points in the upper trapezius muscle (UTM) were randomized to receive DN ($n = 15$), MR ($n = 14$), or sham DN ($n = 15$). The PPT over the UTM (ipsilateral and contralateral sides) and the proximal head of the radius (ipsilateral and contralateral to the treated side) and neck pain were assessed immediately and 10 minutes after the intervention.

Results: There was no significant Group $\times$ Time interaction for PPT in the UTM on the treated side ($F = 0.63$, $P = .641$) or the contralateral side ($F = 1.77$, $P = .144$). However, there was a main effect of time on both the treated side ($F = 4.917$, $P = .001$) and the contralateral side ($F = 4.70$, $P = .015$), with DN and MR increasing PPT at the UTM. No significant Group $\times$ Time $\times$ Side interaction was found for PPT at the proximal head of the radius ($F = 1.23$, $P = .276$). Within-group analysis revealed a significant increase in PPT on the ipsilateral and contralateral sides in both DN and MR. Neck pain decreased after DN ($P < .001$), MR ($P < .001$), and sham DN ($P = .008$).

Conclusion: A single application of DN or MR generated local and distant hypalgesic responses superior to placebo. Future trials are needed to examine whether these findings occur in long-term follow-ups. (J Manipulative Physiol Ther 2021;44:95-102)

Key Indexing Terms: Dry Needling; Musculoskeletal Manipulations; Trigger Points

^a Doctoral Program in Health Sciences, Federal University of Health Sciences of Porto Alegre, Porto Alegre, Brazil.

^b Feevale University, Health Science Institute, Novo Hamburgo, Brazil.

^c Sydney School of Public Health, Faculty of Medicine and Health, The University of Sydney, Sydney, Australia.

^d Institute for Musculoskeletal Health, Sydney Local Health District, Sydney, Australia.

^e Centro Universitário Ritter dos Reis / UniRitter, Porto Alegre, Brasil.

^f Private Practice, Porto Alegre, Brazil.

Corresponding author: Fábio Franciscatto Stieven, 200 Dr Carlos Renato R. Fonseca Street, Apt. 101-B, Porto Alegre, RS 91210120, Brazil.

(e-mail: fabiostieven@feevale.br).

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INTRODUCTION

Myofascial pain syndrome is a common source of musculoskeletal pain and is often associated with the presence of myofascial trigger points (MTrPs), defined as hyperirritable spots within a taut muscle band that are painful to compression and stretch.¹ In conditions of the upper quarter, such as neck pain,² headache,^{3,4} and nonspecific shoulder pain,⁵ the upper trapezius muscle (UTM) is commonly affected by MTrPs. The presence of an MTrP can elicit both local and referred pain, which is recognized as familiar by the patient.⁶⁻⁹ Moreover, snapping palpation or mechanical stimulation of the MTrP by a needle^{1,10} can evoke local twitch responses. It has been demonstrated that an active MTrP contains elevated levels of proinflammatory substances, such as substance P, bradykinin, and

serotonin, thus constituting a sensitive locus that tends to facilitate spinal cord and supraspinal pathways, favoring the phenomenon of central sensitization.¹¹⁻¹³ For instance, Shah et al⁹ describe higher concentrations of inflammatory mediators, neuropeptides, cytokines, and catecholamines in blood samples of individuals presenting with active MTrPs on the UTM compared to those with latent or no MTrPs in the UTM. Nevertheless, altered levels of these substances can also be found in the gastrocnemius muscle, suggesting that the elevation of biochemical substances may not be limited to areas around the muscle affected by the active MTrP.

Among the nonpharmacologic treatment options for myofascial pain syndrome, myofascial release (MR) and dry needling (DN) have been investigated in previous clinical trials.^{8,14-17} Although the former is eminently manual, the latter is instrumental and invasive. Two systematic reviews with meta-analysis^{18,19} have demonstrated potential benefits in pain reduction in favor of DN over sham treatment in short-term and midterm follow-ups. Moreover, a systematic review by Cagnie et al²⁰ shows moderate evidence for ischemic compression and strong evidence for dry needling as effective treatment strategies to reduce local pain in the UTM in individuals with chronic neck pain.

Llamas-Ramos et al¹⁷ compared the effects of DN and manual therapy applied to MTrPs in the UTM and measured the pressure pain threshold (PPT) in the spinous process of C7. A drawback of this study was the absence of a sham group, which precludes drawing adequate conclusions regarding the true treatment effect of both techniques. In this 3-arm randomized controlled trial, we aimed to compare the immediate effects of a single intervention of DN, MR, or sham DN in individuals with chronic neck pain and MTrPs in the UTM unilaterally. Specifically, we sought to determine the effects of these treatments on PPT at the UTM (treated and contralateral sides) and proximal radius head (ipsilateral to the treated side and contralateral side) immediately and 10 minutes after the interventions. The primary hypothesis of this trial was that participants enrolled to the DN group would display a significant increase in local and distant PPT compared to the MR and sham DN groups. Furthermore, as previous studies have shown that manual therapy directed to the cervical spine can immediately elicit an increase in PPT at distant pain-free sites (eg, the elbow)²¹⁻²³—probably because of segmental and central inhibitory mechanisms—our secondary aim was to verify the effects of DN, MR, and sham DN on PPT at the ipsilateral and contralateral proximal head of the radius.

METHODS

The study design was approved by the Ethics and Research Committee of the Federal University of Health Sciences of Porto Alegre (registration number: 12-997) and registered at ReBEC (Registro Brasileiro de Ensaios Clínicos; registration

number: RBR-9p4f7d). This trial was reported according to the recommendations of the Consolidated Standards of Reporting Trials (CONSORT) statement.²⁴

Participants

Adults aged 18 to 50 years with chronic neck pain (ie, symptoms lasting for at least 12 weeks) presenting with an MTrP in only 1 UTM were enrolled from December 2012 to November 2013. The study was conducted at the university's laboratory in Porto Alegre, Brazil. MTrP diagnosis was performed following the criteria described by Simons et al¹: presence of a sensitive palpable nodule within a taut band, presence of local twitch response on snapping palpation, reproduction of a typical pattern of referred pain in response to compression, and recognition of the pain as familiar by the patient. Exclusion criteria were a history of upper quadrant surgery, nonmyofascial disorders, presence of MTrPs bilaterally, complaints of pain or injuries around the elbow or forearm, receipt of treatment for neck pain within the past 12 weeks, and whiplash injury. Participants that met the eligibility criteria provided informed consent and entered the trial.

Study Protocol

The diagnosis of MTrP was performed by a physiotherapist with 6 years of experience in the treatment of myofascial disorders and manual therapy. All procedures were performed in a silent and controlled laboratory (23°C ± 1°C). Before randomization, individuals completed a demographic questionnaire, the Portuguese version of the Neck Disability Index,²⁵ and the Portuguese version of the Fear-Avoidance Beliefs Questionnaire²⁶ to characterize the sample in terms of disability and fear-avoidance beliefs. After baseline examination, an assistant not involved in the recruitment and treatment performed the simple randomization procedure via internet-based randomization software (www.random.org). Allocation concealment was guaranteed by using sealed and opaque envelopes. All participants were informed that they would be randomly assigned to 1 of the 3 groups of the study (DN, MR, or sham DN). A third researcher carried out the single treatment session. All outcome measures were assessed by a trained researcher, blinded to treatment assignment, at 2 time points: immediately after treatment and 10 minutes afterward.

Primary Outcome: PPT

Pain pressure threshold, which is defined as the least amount of pressure applied to a particular point capable of producing a perceived pain response,²⁷ was measured on the UTM bilaterally and in the proximal head of the radius bilaterally. The proximal head of the radius was chosen because of the segmental relationship between neck pain

and possible referred symptoms extending to the forearm, caused by cervical disk disease or lower cervical root irritation, for example. A digital algometer (FPX 25; Wagner Instruments, Greenwich, Connecticut) with a 1-cm² rubber tip was used, which displayed force digitally in increments of 0.1 kg/cm². A trained researcher applied pressure at a constant rate with the device perpendicularly to the selected point. The participant was comfortably seated during the entire process of measurement. The reliability of pressure algometry for the UTM in individuals with neck pain has been found to be high²⁸ (intrarater intraclass correlation coefficient [ICC]=0.96, 95% confidence interval [CI], 0.91-0.98; interrater ICC=0.81, 95% CI, 0.67-0.90). Participants were instructed to verbalize painful responses to the least amount of pressure that elicited pain. The mean of 3 measurements, with an interval of 30 seconds between measurements, was expressed in kg/cm². The PPT assessment was performed before the intervention, immediately afterward, and at a 10-minute follow-up. The minimal detectable change for PPT over the UTM in individuals with neck pain²⁸ is 42.7 kPa (0.435 kg/cm²).

Secondary Outcome: Neck Pain Intensity

Current neck pain was verified by the numerical pain rating scale, an 11-point numerical scale ranging from 0 (no pain) to 10 (worst pain ever experienced). Test-retest reliability was found to be good (ICC=0.76, 95% CI, 0.51-0.87), and the minimal detectable change and minimum clinically important difference are 1.3 and 2.1 points, respectively, in individuals with cervical pain.²⁹ Neck pain was assessed at baseline and at the 10-minute follow-up.

Intervention

Dry Needling. Dry needling was applied unilaterally in the active MTrP located in the UTM. Sterile stainless-steel acupuncture needles (25 × 0.3 mm; Dongbang, Seoul, Korea) were used. The needle was introduced subcutaneously, penetrating the skin to a depth of 10 to 15 mm. The needle was manipulated until a local twitch response was elicited. Once the first local twitch response was obtained, vertical pistoning of the needle was performed so that a local twitch response was elicited at least 3 more times. No rotational movement was performed during treatment. The needle was left on the MTrP for approximately 10 seconds and then removed.

Myofascial Release. With the UTM in a lengthened position, gentle pressure was applied to the MTrP until the sensation of pressure became painful. The pressure was then sustained until the pain was reduced by about 50%. An increment in pressure until the onset of pain was performed 2 more times. The technique was applied with the participant comfortably seated on a chair with the back supported and the upper limb on the treatment side hanging beside

the body to relax the periscapular muscles. The treating therapist used his thumb and index fingers to palpate the UTM using a pincer palpation. Contact was made with both hands. In this position, MR was performed until a gentle release of the taut band was perceived by the therapist. Finally, the UTM was passively stretched for 30 seconds. The combination of MR plus stretching was performed 3 times, with an average duration of 90 seconds.

Sham DN. Participants in this group were comfortably placed in a prone position. The therapist manipulated the tissues adjacent to the MTrP on the symptomatic side for 10 seconds, with a light touch on the skin, using his thumb and index finger. Then the needle guide tube was positioned on the MTrP, where a gentle tap was performed. No needle puncture occurred.

Adverse Events

Participants were asked to report any adverse symptoms that they experienced immediately after intervention and during the 72 hours after the intervention. Adverse events were defined as any symptom perceived as distressing by the participant and requiring further treatment.³⁰ Adverse events were self-reported by the participants, and this information was collected by a clinician not involved in the study.

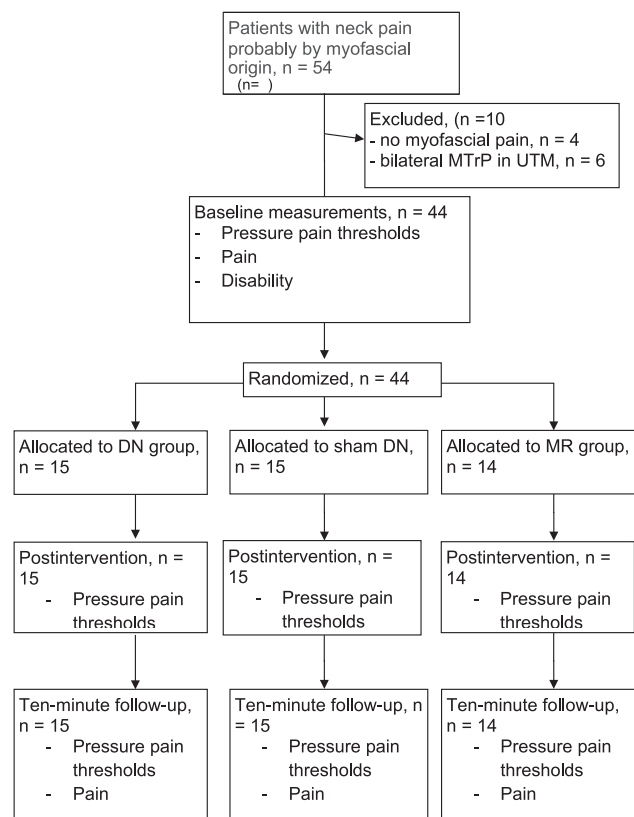


Fig 1. Flow diagram of participants throughout the course of the study.

Sample-Size Determination

Sample size was determined with G*Power version 3.1.6 software.³¹ Sample size was determined a priori considering a between-groups difference of 2.1 points in neck pain intensity, assuming a standard deviation of 2.1, a significance level of 5% ($\alpha = 0.05$), and a power of 90%. In this scenario, 27 participants were required.

Statistical Analysis

Data were analyzed with SPSS 20.0 (SPSS Inc, Chicago, Illinois). Data are expressed as mean $\pm$ standard deviation (SD) and 95% CI. Data normality was assessed by means of the Kolmogorov-Smirnov test, considering $P > .05$. For sex comparison, the Fischer exact test was used.

Baseline features were compared among the 3 groups using 1-way analysis of variance (ANOVA) with Bonferroni correction for multiple comparisons. Because the intervention was unilateral (only in the symptomatic UTM), the proximal head of the radius was classified as ipsilateral (ipsilateral to the treated UTM) or contralateral (contralateral to the treated UTM). For PPT, a $3 \times 3 \times 2$ mixed-model ANOVA with time (baseline, immediately after intervention, and 10-minute follow-up), group (DN, MR,

and sham DN), and side (ipsilateral and contralateral to the treated side) was used to examine the intervention effects on PPT. For neck pain, a 2×3 mixed-model ANOVA with time (baseline and 10-minute follow-up) and group (DN, MR, and sham DN) was used. The hypothesis of interest was the Group $\times$ Time interaction, with Bonferroni correction used for the posttest comparisons. Within-group effect sizes were calculated using Cohen's d . An effect size greater than 0.8 was considered large; 0.5, to 0.8 was considered moderate; values below 0.5 were considered small.

RESULTS

Fifty-four individuals with neck pain were screened for possible eligibility criteria. Forty-four of these individuals (mean $\pm$ SD age, 27 ± 6 years; 66% female) satisfied the eligibility criteria, gave informed consent, and were randomized to the DN group ($n = 15$), the MR group ($n = 14$), or the sham DN group ($n = 15$). Reasons for ineligibility are presented in Figure 1. There were no dropouts during the study protocol. Baseline demographic and PPT data were similar among the 3 groups (Table 1). No adverse effects were reported in any group.

Table 1. Baseline Demographic Characteristics of the 3 Experimental Groups

Characteristic	DN (n = 15)	MR (n = 14)	Sham DN (n = 15)	P
Sex, n (%)				.919
Male	5 (33.3)	4 (28.6)	4 (27.7)	
Female	10 (66.7)	10 (71.4)	11 (73.3)	
Age, y	27.1 ± 4.7	26.6 ± 3.9	28.2 ± 9.4	.794
BMI, kg/m ²	22.8 ± 3.3	24.0 ± 2.8	23.3 ± 3.8	.652
Duration, min	31.8 ± 14.9	33.6 ± 17.1	30.8 ± 18.2	.906
Neck pain (NPRS)	4.50 ± 1.18	4.79 ± 1.19	4.71 ± 1.33	.809
NDI	12.4 ± 5.2	10.4 ± 3.2	12.6 ± 2.8	.272
FABQ-PA	7.4 ± 5.1	8.9 ± 5.8	9.1 ± 6.2	.672
FABQ-W	12.2 ± 8.7	10.2 ± 6.4	11.1 ± 8.9	.810
Pain pressure threshold, kgf				
Treated UTM (MTrP)	1.46 ± 0.31	1.38 ± 0.38	1.41 ± 0.51	.868
Untreated UTM (asymptomatic)	1.44 ± 0.49	1.35 ± 0.30	1.39 ± 0.93	.927
Proximal head of ipsilateral radius	1.47 ± 0.26	1.47 ± 0.54	1.47 ± 0.66	.999
Proximal head of contralateral radius	1.48 ± 0.38	1.44 ± 0.48	1.42 ± 0.71	.960

Data are presented as mean $\pm$ SD, except for sex.

BMI, body mass index; DN, dry needling; FABQ-PA, Fear-Avoidance Beliefs Questionnaire—Physical Activity subscale; FABQ-W, Fear-Avoidance Beliefs Questionnaire—Work subscale; MR, myofascial release; MTrP, myofascial trigger point; NDI, Neck Disability Index; NPRS, 11-point numerical pain rating scale; UTM, upper trapezius muscle.

Values for PPT are outlined in Table 2. The $3 \times 3 \times 2$ mixed-model ANOVA did not indicate significant Group $\times$ Time $\times$ Side interactions for treated UTMs ($F = 0.63$, $P = .641$) or untreated UTMs ($F = 1.77$, $P = .144$). However, there was a main effect of time in both treated ($F = 4.917$, $P = .001$) and untreated ($F = 4.70$, $P = .015$) sides, with DN and MR groups experiencing an increase in PPT in the UTMs bilaterally; there was no significant difference over time in the sham DN group ($F = 2.471$, $P = .678$). In the untreated UTMs, a significant Group $\times$ Time interaction was seen for PPT ($F = 3.045$, $P = .040$); those who received DN showed a significant increase in PPT compared to MR and sham DN ($P = .204$ and .866, respectively).

At the proximal head of the radius, there were no significant Group $\times$ Time $\times$ Side ($F = 0.0652$, $P = .276$), Side $\times$ Group ($F = 1.379$, $P = .426$), or Group $\times$ Time interactions ($F = 1.728$, $P = .298$), nor main effects of group ($F = 3.228$, $P = .667$) or time ($F = 2.195$, $P = .855$). In the within-group analysis, however, the DN and MR groups showed significant increases in PPT in the ipsilateral

proximal head of the radius ($P = .004$ and .018, respectively). There was no significant difference over time for the sham DN group ($P = .917$). For the proximal head of the radius in the contralateral side, only the DN group showed a significant increase in PPT ($P = .037$; MR and sham DN, respectively; $P = .077$ and $P = .972$). Within-group effect sizes were large in the DN group (1.64 for UTM and 0.90 for the proximal head of the radius of the treated side) and small to moderate in the MR group (0.62 for UTM and 0.26 for the proximal head of the radius of the treated side) (Table 3).

For neck pain, the 2×3 mixed-model ANOVA showed significant reduction in pain scores immediately after the intervention (Table 4) for all 3 groups (DN: $P < .001$; MR: $P < .001$; sham DN: $P = .008$). However, only DN and MR had a clinically significant reduction in differences between average pain scores (DN: 2.50 ± 1.29 ; MR: 2.64 ± 1.33 ; sham DN: 0.41 ± 0.13).

DISCUSSION

The results of this 3-arm randomized clinical trial study provide evidence that a single session of DN and MR, but not sham DN, results in an increase in PPT at the UTM bilaterally and at the ipsilateral proximal head of the radius. The PPT at the contralateral proximal head of the radius also increased after DN. We also noted that neck pain intensity significantly decreased in both DN and MR groups, clinically superior to the placebo. Participants treated with DN revealed an important size effect, suggesting a strong clinical effect, whereas those who received MR presented a small to moderate magnitude.

Recent systematic reviews with meta-analysis have found DN effective for reducing pain immediately after treatment and in short- and midterm follow-ups compared to sham or control groups,^{18,19} but provide mixed conclusions regarding the effects of DN compared to other treatments.^{19,20} Our study contributes to the current body of evidence by demonstrating that both DN and MR produce comparable results in terms of increase in PPT and reduction in neck pain intensity. A clinically important reduction in neck pain intensity after DN and MR but not sham DN was found, corroborating the findings of previous systematic reviews.

After DN and MR, an increase in PPT occurred at the ipsilateral proximal head of the radius—and an increase as well in PPT at the contralateral proximal head of the radius after DN—which indicates that DN and MR may produce hypalgesic effects in pain-free points. This finding is in line with those of a previous trial¹⁶ demonstrating significant increase in PPT in the UTM and decrease in neck pain intensity immediately after a single application of DN in an MTrp located in the extensor carpi radialis longus muscle. It has been proposed that the observed remote effect could

Table 2. Outcome Data for Pressure Pain Threshold (kgf)

Group	Baseline	Postintervention	10-Minute Follow-up
Treated UTM			
DN	1.47 $\pm$ 0.31	2.39 $\pm$ 0.81 ^a	2.57 $\pm$ 0.83 ^a
MR	1.38 $\pm$ 0.37	1.72 $\pm$ 0.58 ^a	1.77 $\pm$ 0.51 ^a
Sham DN	1.38 $\pm$ 0.76	1.49 $\pm$ 0.96	1.58 $\pm$ 0.98
Untreated UTM			
DN	1.45 $\pm$ 0.49	1.97 $\pm$ 0.99 ^a	2.08 $\pm$ 0.81 ^a
MR	1.35 $\pm$ 0.31	1.51 $\pm$ 0.44 ^a	1.54 $\pm$ 0.32 ^a
Sham DN	1.39 $\pm$ 0.93	1.42 $\pm$ 0.86	1.43 $\pm$ 0.92
Ipsilateral radius			
DN	1.47 $\pm$ 0.26	1.79 $\pm$ 0.45 ^a	1.87 $\pm$ 0.8 ^a
MR	1.48 $\pm$ 0.54	1.63 $\pm$ 0.62 ^a	1.68 $\pm$ 0.52
Sham DN	1.47 $\pm$ 0.66	1.47 $\pm$ 0.82	1.51 $\pm$ 0.81
Contralateral radius			
DN	1.48 $\pm$ 0.39	1.67 $\pm$ 0.44 ^a	1.77 $\pm$ 0.59
MR	1.44 $\pm$ 0.48	1.54 $\pm$ 0.50	1.64 $\pm$ 0.32
Sham DN	1.43 $\pm$ 0.71	1.42 $\pm$ 0.84	1.40 $\pm$ 0.94

Data are presented as mean $\pm$ SD.

DN, dry needling; MR, myofascial release; UTM, upper trapezius muscle.

^a Statistically significant difference from baseline ($P < .05$).

Table 3. Between-Groups Differences (95% CI) in Pressure Pain Threshold (kgf)

Time	DN vs Sham DN	DN vs MR	MR vs Sham DN
Treated UTM (MTrP)			
Postintervention	0.90 (0.81-0.98) ^a	0.66 (0.79-0.54) ^a	0.23 (0.03-0.44) ^a
10-min follow-up	1.00 (0.92-1.08) ^a	0.81 (0.64-0.98) ^a	0.19 (0.07-0.31) ^a
Untreated UTM			
Postintervention	0.55 (0.48-0.63) ^a	0.47 (0.18-0.76) ^a	0.09 (−0.02 to 0.20)
10-min follow-up	0.66 (0.60-0.72) ^a	0.55 (0.29-0.80) ^a	0.11 (0.01-0.21) ^a
Ipsilateral radius			
Postintervention	0.32 (0.13-0.52) ^a	0.16 (0.08-0.25) ^a	0.16 (0.05-0.27) ^a
10-min follow-up	0.36 (0.18-0.53) ^a	0.19 (0.17-0.21) ^a	0.17 (0.06-0.28) ^a
Contralateral radius			
Postintervention	0.13 (0.05-0.21) ^a	0.26 (0.07-0.44) ^a	0.13 (0.10-0.16) ^a
10-min follow-up	0.37 (0.19-0.55) ^a	0.13 (0.05-0.21) ^a	0.24 (0.11-0.36) ^a

Data are presented as mean (95% confidence interval). Positive scores are in favor of the reference group (the first group listed in the comparison).

DN, dry needling; MR, myofascial release; MTrP, myofascial trigger point; UTM, upper trapezius muscle.

^a Statistically significant ($P < .05$).

be mediated by pain modulation at the spinal level, probably owing to the diffuse noxious inhibitory control mechanism.³² Borg-Stein and Simons⁶ argue that local pressure over a taut band would equalize the length of the sarcomere

Table 4. Outcome Data for Neck Pain Intensity

Group	Baseline	Postintervention
DN ^a	4.50 ± 1.18	2.00 ± 1.07
MR ^a	4.79 ± 1.19	2.14 ± 1.03
Sham DN ^a	4.71 ± 1.33	4.29 ± 1.44
Within-group change from baseline ^b		
DN		2.5 (1.8-3.2) ^c
MR		2.6 (1.9-3.4) ^c
Sham DN		0.4 (0.1-0.7) ^c
Between-groups difference in change (vs sham DN) ^b		
DN		2.1 (1.1-3.1) ^c
MR		2.2 (1.2-3.2) ^c

DN, dry needling; MR, myofascial release.

^a Data are presented as mean ± SD on a 0-to-10 scale.

^b Data are presented as mean (95% confidence interval).

^c Statistically significant ($P < .01$).

involved, thereby decreasing the feedback cycle that composes the integrated hypothesis of myofascial pain perpetuation. It is possible that both techniques act on both segmental and central inhibitory mechanisms.^{33,34} Despite the plausible role of local effects due to stimulation of large-diameter, low-threshold mechanoreceptors at the spinal-cord level,³⁵ activation of descending inhibitory pathways may play a role in the observed increase in local and remote PPT.³⁶⁻³⁸ However, the exact mechanism of DN and MR is not entirely known, and whether they act by the same mechanism of action remains speculative.^{33,34} Future studies should address potential mechanisms of both therapies.

It has been argued that deep stimulation of an MTrP could produce adverse effects such as posttreatment local and referred pain.³⁹ To overcome this common adverse effect, a recent trial has shown that ischemic compression after DN reduces posttreatment soreness and increases cervical range of motion.⁴⁰ In our trial, no adverse events were noted immediately after treatment or within 72 hours afterward. This finding is interesting, as a local twitch response was generated in all participants allocated to the DN group.

Limitations

Our clinical trial has some potential limitations. First, it focused on only the immediate response after treatment.

Trials with long-term follow-ups might provide different results. A single clinician performed all treatments, which may have introduced selection bias because of clinician preferences. The likelihood of selection bias was reduced, as the proposed treatments were highly standardized. Third, monitoring of the treatment effect on range of motion and other important participant-reported outcomes was not performed. And fourth, we did not verify participant expectations regarding treatment outcomes with DN and MR, and we did not verify participants' awareness regarding the treatment received if it was not a validated sham procedure. Future studies must include a greater number of clinicians performing treatments. Furthermore, a higher number of treatment sessions and other associated outcomes must be included.

CONCLUSION

The application of a single intervention with DN or MR produced clinical positive effects in immediate response and at 10-minute follow-up. Dry needling produced an increase in PPT for local pain of an MTrP over the UTM and distant pain-free regions (head of radius) compared to MR. The effect of DN was stronger than and superior to that of MR, which showed a similar response as sham treatment in PPT levels at pain-free regions. Dry needling was similar to MR in reducing self-reported pain.

FUNDING SOURCES AND CONFLICTS OF INTEREST

No funding sources or conflicts of interest were reported for this study.

CONTRIBUTORSHIP INFORMATION

Concept development (provided idea for the research): F.F.S., G.E.F., M.F.S., L.H.T.R.

Design (planned the methods to generate the results): F.F.S., G.E.F., R.F.A.

Supervision (provided oversight, responsible for organization and implementation, writing of the manuscript): F.F.S., F.X.A., R.F.A.

Data collection/processing (responsible for experiments, patient management, organization, or reporting data): F.F.S., F.X.A.

Analysis/interpretation (responsible for statistical analysis, evaluation, and presentation of the results): G.E.F., M.F.S., L.H.T.R.

Literature search (performed the literature search): F.F.S., R.F.A.

Writing (responsible for writing a substantive part of the manuscript): F.F.S., G.E.F., F.X.A., M.F.S., L.H.T.R.

Critical review (revised manuscript for intellectual content, this does not relate to spelling and grammar checking): F.F.S., M.F.S., L.H.T.R.

Practical Applications

- A single application of dry needling generated local and distant hypalgesic responses superior to placebo.
- A single application of myofascial release generated local and distant hypalgesic responses superior to placebo.
- Dry needling promoted hypalgesic effects greater than those of myofascial release.

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